VS Code & Python – Software Requirements

Pre-training setup guide | Version 1.1 – Naji El Kotob

> ***Purpose:*** *Get a working Python and VS Code environment on your laptop before the training starts. It takes about 15 minutes.*

General requirements

- A laptop with Windows 10 or 11 and a stable internet connection
- Admin rights to install software – on a company laptop, ask your IT team before the training
- About 1 GB of free disk space

Step 1 – Install Python

1. Go to python.org/downloads and download the latest **Python 3** Windows installer
2. Run the installer and tick **Add python.exe to PATH** at the bottom of the first screen
3. Click **Install Now** and wait until you see "Setup was successful"

Why PATH matters: without it, Windows and VS Code can't find Python, and you get the error *'python' is not recognized*. It's the most common setup mistake.

Step 2 – Install VS Code

1. Go to code.visualstudio.com and click **Download for Windows**
2. Run the installer and keep the default options. Make sure Add to PATH is selected.
3. Optional: tick **Add "Open with Code" action** to open folders from File Explorer with a right-click

Step 3 – Install the Python extension (required)

1. Open VS Code and click the **Extensions** icon in the left bar (or press `Ctrl+Shift+X`)
2. Search for **Python** and install the one published by **Microsoft**

Why it's required: VS Code is just an editor – this extension lets it find your Python, run and debug your code, and suggest code as you type. It also installs Pylance and the Python Debugger automatically, so this is the only extension you need.

Step 4 – Verify the setup

1. Create a folder for the training, for example `C:\python-training`, and open it in VS Code (**File > Open Folder**)
2. Open the terminal (**Terminal > New Terminal**) and run:

```
python --version
```

You should see something like `Python 3.x.x`.

3. Create a file named `hello.py` with this line:

```
print("Hello, Python!")
```

4. Click the ▶ **Run** button at the top right – `Hello, Python!` appears in the terminal

If VS Code asks you to select an interpreter, press `Ctrl+Shift+P`, type **Python: Select Interpreter** and pick the Python you installed in Step 1.

Step 5 – Virtual environment (optional)

A virtual environment keeps the packages for this training in its own folder, so they don't clash with other Python projects on your laptop. Skip this step if you're new to Python – the trainer will cover it in the sessions.

In the VS Code terminal, inside your training folder, run:

```
python -m venv .venv
.venv\Scripts\activate
```

`(.venv)` appears at the start of the terminal line when it's active. Install packages with `pip install <package-name>`.

If PowerShell blocks the activate script, run this once, then try again:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

Troubleshooting

Problem	Fix
'python' is not recognized	PATH wasn't ticked – run the Python installer again, choose **Modify**, then tick **Add Python to environment variables**
Typing `python` opens the Microsoft Store	Turn off the Store shortcut: **Settings > Apps > Advanced app settings > App execution aliases**, switch off both Python entries
VS Code doesn't see Python	Close and reopen VS Code after installing Python, then use **Python: Select Interpreter**
The installer asks for an admin password	Ask your IT team to install Python and VS Code, or to grant temporary admin rights

References

- Python downloads
- VS Code download
- Python in VS Code – getting started
- Python virtual environments (venv)